

> "Cortisol dysfunction can lead to **inflammation and reduced bone formation, causing joint pain** or even osteoarthritis"
>
> **–DR SUNIL DACHEPALLI,** *Orthopaedic surgeon, Yashoda Hospitals, Hyderabad*



hormone (ACTH), to the triangle-shaped adrenal glands located above the kidneys. This is the real centre of stress management. On receipt of the emergency signal, the adrenal glands send out reinforcements in the form of cortisol. This steroidal hormone, in turn, activates the body's sympathetic nervous system, causing the heart to race, the pulse and blood pressure to shoot up, and the breath to become faster. More oxygen and glucose are released in the brain and the bloodstream, providing the body the burst of energy it needs to combat stress. Once the danger is past, cortisol levels drop and the body returns to normal.

The problem arises when stress becomes a chronic condition and there is too much cortisol in the body, and the slightest of stressful situations, say, a traffic jam, triggers its secretion. "Cortisol," says Dr Mishra, "is our survival hormone and is controlled by our thoughts. When our thoughts turn stressful, the body releases this hormone to actually protect itself. When we are stressed for long periods of time, there can be too much cortisol in the body, which is not good for either our health or our well-being."

Think of it as a malfunction of the body's thermostat, which continues to keep the body heated. That is what excess chemicals to combat everyday stress do. "When cortisol was first discovered," says Dr P.N. Renjen, a Delhi-based neurologist, "it was a breakthrough. But we soon realised that steroids can be a double-edged sword. Have too much in the body and they start doing the opposite of what they are meant to do—that is, protect the body—causing inflammation. Excess cortisol has now been linked to permanent changes in the brain and is thus a precursor to several neurological diseases." Studies confirm that the stress response in recent years is being activated more frequently than it should. For example, a July 2018 study by the Washington D.C.-based Society for Research in Child Development found that a drop in high school grades increased the cortisol levels in students. Another study by Oracle and US research firm Workplace Intelligence in October 2021 of 41,600 professionals in 13 countries found Indians to be the most stressed—with 91 per cent of the 1,100 respondents saying they were highly stressed compared to the global average of 80 per cent. The stressors varied from financial worry to emotional issues to career-related problems to disconnect with their own lives. "Our body's stress response is geared for life-or-death situations," says Dr Samir Parikh, director of mental health and behavioural sciences at Fortis Healthcare. "But today that mechanism can get triggered for something as trivial as getting late to office and worrying about your boss yelling at you or simply forgetting someone's birthday or reading something online." Overthinking and overanalysing have become the bane of our existence and the reasons for our heightened levels of stress. "Your thoughts control your brain's perception of stress. If you are constantly fearing the worst, you are unconsciously sending the wrong signal to the hypothalamus," Dr Parikh adds.

CAUSE OF ALL ILLS

The World Health Organization had in 2016 dubbed stress "the health epidemic of the 21st century". While stress may not be a disease with specific causes or symptoms, it wreaks sufficient havoc on the body's organs over time to become life-threatening. "In medical terms," says Dr Atul Ingale, senior nephrologist at Fortis Hospital, Mulund, Mumbai, "we view stress as a chronic inflammatory state of the body." Talking about how it affects kidneys, he says the impact is not direct but how it affects the lifestyle—causing people to drink less water, compromise on nutrition and turn more prone to diabetes or hypertension, all known causes of chronic kidney disease. Diabetes, likewise, is linked to stress. A 2017 study in the journal *PLOS One* found stress to be a risk factor for Type